

The Chair of Operations Management, in cooperation with **Realog Consulting GmbH**, is offering a

Project Study / IDP

on the topic

From Analog to Digital — Building the Capacity & KPI Analytics Platform for Air Cargo Terminals

open to

Bachelor Students and Master Students of the TUM School of Management and Master Students of the TUM School of Computation, Information and Technology

About Realog Consulting

Realog has supported airports, ground handlers, airlines and logistics companies worldwide since 2003, covering the entire lifecycle of air cargo hub projects, from analytics, capacity forecasting and master planning through design and tendering to IT implementation. Landmark projects include SATS, WFS, Swissport, Lufthansa Cargo, Emirates SkyCargo, Adani, Royal Jordanian, Celebi and GMR, including fully automated cargo terminals.

Scope

Air cargo terminal design and operations at Realog rely on extensive capacity and performance analyses. Currently, this data originates from complex, project-specific Excel workbooks and requires structured cleansing, validation, and harmonization before it can be used for systematic benchmarking, KPI steering, or advanced analytics.

Within this project, a student team will:

- Clean, validate, and harmonize heterogeneous project and operational data (cargo volumes, flight and truck data, shipment structures) into a single, governed database.
- Implement the capacity calculation engine: flow-specific throughput, the peak-factor cascade, and derivation of functional capacity demand.
- Enable scenario management to define, store, and compare design scenarios side by side.
- Build a benchmarking module to compare key indicators across projects, terminals, and regions.
- Develop an operational KPI dashboard providing full visibility over the defined KPI set, from executive view down to operational detail.
- Automate executive KPI reporting, including data-driven recommendations, and quantify optimization levers in operational and financial terms.

The main outcome will be:

1. A clean and validated capacity and performance database.
2. A working analytics platform with KPI dashboard and automated executive reporting.
3. Quantified optimization levers and recommendations for management action.

This project transforms Realog's analog, Excel-based analysis landscape into a digital platform, building the foundation for future benchmarking, forecasting, and AI-supported optimization initiatives.

The team's profile

- Strong interest in aviation, airline operations, and data-driven decision making
- Proficiency in Python
- Experience with data cleaning, preprocessing, and working with real-world datasets
- Strong analytical and problem-solving skills
- Reliable, self-driven and structured working style
- Ability to work independently in a fully remote **team** environment
- **Bonus:** Knowledge in Air Cargo and Operations

Start and duration: As soon as possible; duration: 3-4 months full-time, option of subsequent internship or part-time work

Language: English

Number of participants: 2-4

Application Requirements: Applicants may apply either as a pre-formed team or individually. Individual applicants will be assigned to a team or integrated into an existing team by the chair in coordination with the corporate partner after the selection process.

Application Documents

- CV
- Master candidates: Full Transcript of Records of Bachelor and Master
- Bachelor candidates: Full Transcript of Records of the Bachelor

Supervisor TUM: Aziz Boushih, Research Assistant, Chair of Operations Management (Prof. Kolisch)

Supervisor Realog Consulting: Katharina Kiesler, Head of Business Development

Contact: Applications, please to aziz.boushih@tum.de